

# ABDULLA ZOHAIB HK

AI/ML & Software Engineering Intern | Python | AI Systems | Backend Development

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## PROFESSIONAL SUMMARY

Computer Science engineering student specializing in AI & ML with hands-on experience building **AI-powered software systems, simulation platforms, cloud observability tools, and interactive 3D applications**. Strongest in Python/FastAPI backend development, AI workflow integration, REST APIs, databases, Docker, and end-to-end product implementation.

## TECHNICAL SKILLS

|               |                                                                                             |
|---------------|---------------------------------------------------------------------------------------------|
| Languages     | Python, Java, C++                                                                           |
| AI / ML       | Machine Learning, Deep Learning, AI Systems, LLM Integration, LangGraph, TensorFlow, Pandas |
| Backend       | FastAPI, REST APIs, SQLAlchemy, Pydantic                                                    |
| Frontend / 3D | React, Three.js, React Three Fiber                                                          |
| Data / Cloud  | PostgreSQL, SQL, Docker, Docker Compose, Kubernetes                                         |
| Observability | Prometheus, Grafana                                                                         |
| Tools         | Git, GitHub, VS Code, Ollama                                                                |

## SELECTED PROJECTS

### TwinMind AI | AI-Powered Warehouse Digital Twin & Decision Simulation Platform

- Built an interactive 3D warehouse Digital Twin with React, Three.js, and React Three Fiber to visualize receiving, storage, picking, packing, shipping, workers, equipment, inventory, and operational flow.
- Implemented scenario modeling, simulation metrics, baseline-vs-what-if comparison, optimization, AI-assisted recommendations, validation, and human-in-the-loop decision support using FastAPI and PostgreSQL.

### CloudSentinel AI | AI-Powered Cloud Incident Response & Observability Platform

- Developed a FastAPI platform for incident management, severity classification, AI-assisted root-cause analysis, recovery recommendations, and operational health monitoring.
- Integrated LangGraph, Ollama, Prometheus, Grafana, Docker, Kubernetes, SQLite, and automated backend testing to connect AI reasoning with cloud-native observability workflows.

### HealOps AI | AI-Powered Incident Response & Observability

- Built an AI-driven workflow that analyzes logs, detects potential anomalies, performs root-cause analysis, and produces actionable recovery recommendations through REST APIs.
- Worked with FastAPI, LangGraph, Ollama, Docker, Kubernetes, Prometheus, and Grafana to combine local LLM reasoning with application health and infrastructure monitoring.

### Apex AI | AI Response Comparison Platform

- Built a platform that compares responses from multiple AI models and produces a synthesized result using scoring-based evaluation to support faster answer selection.
- Implemented the backend workflow with Python and FastAPI, focusing on API integration, response normalization, comparison logic, and a user-facing result flow.

## EDUCATION

### B.Tech — Computer Science & Engineering (AI & ML) | Anjuman Institute of Technology and Management

Relevant focus: Machine Learning, Deep Learning, Data Analysis, AI Systems, Python, SQL, Software Development

## CORE STRENGTHS

AI application development • Backend engineering • REST API design • Simulation & optimization • Cloud observability • 3D visualization • End-to-end project delivery